

The empirical closed form for OEIS A199833, proved from an exact model

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A199833 carries an empirical closed form: a polynomial in n of degree 3, recorded on the entry and never marked settled. It is true. The entry's count is given exactly by an Ehrhart quasi-polynomial, from which a provably satisfies a monic linear recurrence of order $S = 5$ derived below; a polynomial of degree 3 satisfies $(z - 1)^{3+1}$; so the difference satisfies the product, and $5 + 3 + 1$ consecutive agreements settle it. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A199833 is “Number of $n \times n$ arrays of 4 elements with zero sum and no two neighbors summing to zero.”. It has offset 1 and begins

$$4, 40, 140, 336, 660, 1144, 1820, 2720, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = (16/3)n^3 - (4/3)n = 4 \cdot A000447(n).$$

As of the “Last modified” line on the live entry (Jul 14 2022 09:59:07, revision 18) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a quasi-polynomial

The array has a FIXED number L of entries and an alphabet that grows with n . Every condition the entry imposes is a boolean combination of statements $\sum_i c_i x_i + c n \bowtie 0$ with integer coefficients and $\bowtie$ one of $=, \neq, <, \leq, >, \geq$: the bounds $|x_i| \leq n$ or $0 \leq x_i \leq n$, a vanishing sum, a difference that must not vanish, a pair that must total exactly n . Each is homogeneous in (x, n) jointly, so the admissible x for a given n are precisely the integer points at height n of a finite union of relatively open rational cones in $\mathbb{R}^{L+1}$, and $a(n)$ is the Ehrhart quasi-polynomial of that union: of degree at most L , with period dividing the heights of the cones' ray generators.

3 The bound

The period is derived, not assumed. Every ray of every cell of the arrangement is cut out by L linearly independent hyperplanes taken from the constraint set; writing that $L \times (L + 1)$ system as M , its null direction has t -component $\pm \det$ of the x -parts, so the primitive generator (x^*, t^*)

has t^* dividing that determinant. Let T be the set of those determinants. A simplicial cone in a triangulation has at most $L + 1$ rays, each contributing a factor $1 - z^{t_i}$ with $t_i \in T$, so

$$A(z) = \prod_{d|t \text{ for some } t \in T} \Phi_d(z)^{L+1}$$

annihilates a , and $S = \deg A = 5$. A congruence condition modulo m is counted one residue class at a time; a class is a coset of $m\mathbb{Z}^L$, and a generator primitive in $\mathbb{Z}^{L+1}$ must be scaled by a divisor of m to lie in it, so each t is replaced by mt and nothing else changes.

Over the common denominator the numerator has degree below that of A for every cell of the arrangement that HAS a ray, because such a cell's numerator collects a fundamental parallelepiped whose heights are below the sum of its generators' heights. The origin is a cell too and it has none: its series is the constant 1, which over A is A/A and reaches degree exactly $\deg A$. The bound in force is therefore $zA(z)$, of order $S = 5$, and $a(0), \dots, a(S-1)$, computed exactly, determine every later term. Nothing is fitted, and the annihilator is tested on terms the model was not asked for before anything is claimed — a bound one too small fails that test, which is how this one was found.

4 The proof

Theorem 1. *The empirical closed form holds for every n .*

Proof. Write P for the claimed polynomial and $r(n) = a(n) - P(n)$. The annihilator A derived above kills a , and $(z-1)^{3+1}$ kills P , so $A(z)(z-1)^{3+1}$ — monic, of order $5 + 3 + 1$ — kills r . The model was evaluated exactly in integer arithmetic and P in exact rational arithmetic at every index from $n = 1$ to $n = 1 + 5 + 3 + 1$, and r vanished at all of them. A monic recurrence with $5 + 3 + 1$ consecutive zeros has every later term zero, so $r \equiv 0$. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 36 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the claimed polynomial was evaluated directly on those published terms, in exact rational arithmetic and with no model involved, and agrees with every one of them.

Third, the derived annihilator was itself tested on terms the model was not asked for: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.